

Shreya Akurathi

+91-9766014952 | sakurath.office@gmail.com | GitHub: [shreyaakurathi](#) | LinkedIn: [shreyaakurathi](#)

EDUCATION

Manipal Institute of Technology | B.Tech, Data Science with Minor in Finance & Risk Analytics | **CGPA**: 8.67 / 10

2020 – 2024

Relevant Coursework: Deep Learning, ML, Multivariate Analysis, NLP, Data Analytics, Finance & Econometrics, DBMS, Statistics

WORK EXPERIENCE

BlackRock | Data Engineering Analyst | Mumbai, India

Jul 2024 - Present

- Engineered a metadata-driven Snowflake loader enabling bidirectional data migration between Snowflake DB and the data lake (Hive & Iceberg), achieving 100% rollout success with integrated role-based access controls.
- Architected config-driven transformations at the Ingestor server level to consume 100% of real-time security level events and created transformed Hive views and lineage timeline visualizations used as primary sources of truth over 3 teams.
- Improved Tableau board latency by 98% (T-2 days → 2 hrs) through frequency optimization and development of a metadata-driven ingestion framework, reducing SLA time by 6 hours and ensuring data integrity with deduplication logic.
- Optimized 20+ Spark datasets via partition tuning, filter pruning, and write-mode optimization, resulting in 90% resource reduction per dataset and 30–45% aggregate decrease in cluster utilization during cloud pipeline transition while ensuring zero data loss.
- Reduced infrastructure failures by 76% as Historical Pipeline SME by implementing proactive monitoring, debugging job failures, and enforcing metadata development standards across two teams.
- Automated CI/CD validation scripts, cutting PR review effort by 90%, and developed a Splunk-based job tracker that enabled real-time alerts and eliminated T-1 latency in job failure detection.

BlackRock | Data Engineering Intern | Mumbai, India

Jan 2024 - Jun 2024

- Reduced system boot-up time by 94% (7+ hrs → <10 mins) by developing a framework to migrate 13K+ scheduled tasks into 4 optimized scheduling cycles, shortening the production lifecycle for client risk analytics by one hour.
- Increased system reliability by implementing automated monitoring for 15 critical settings with 100% error detection accuracy, cutting manual intervention by 80% and production issues by 20% through a scalable, config-driven alert system.
- Improved model accuracy to 97% by building a GPT-4-powered copilot for querying curated datasets and optimizing a RAG architecture alongside PyArrow-based automated data loading for 9 datasets.

Intel Unnati Summer Program | Intern – Deep Learning and Computer Vision | Pune, India

May 2023 – July 2023

- Optimized YOLOv5 model inference using OpenVINO for real-time video processing, achieving a 40% reduction in inference time and a 60% improvement in throughput for real-time social distancing detection.
- Conducted experiments across CPU configurations, achieving 70% average accuracy improvement through performance optimization

Department of Data Science and Computer Applications, MIT Manipal | Undergraduate Researcher | Manipal, India

Oct 2022 – Sep 2023

- Developed channel attention modules & deep learning models for activity recognition & context-aware navigation for student project “Smart Collared Eye”, achieving an 80% improvement in prediction accuracy

Navana Tech | Computational Linguist Intern | Mumbai, India

Dec 2022 – Jan 2023

- Established baseline performance metrics for data validation by preprocessing 2TB+ of audio data using TorchAudio & conducting systematic experiments on the existing speaker verification system.
- Achieved 83% accuracy in multilingual speaker validation and extended the system to Marathi & Kannada, covering five dialects each.

The Disruptive Dispatch | Founder / Research Head | Manipal, India

Sep 2021 – Dec 2022

- Founded a student-led economics & finance newsletter, building a community of 150+ Substack subscribers & 400+ members across Discord & WhatsApp, & authored 10+ essays & led editorial & design initiatives to strengthen reader engagement & brand presence.

The Girl Code | Intern | Pune, India

Jul 2021 – Nov 2021

- Organized campus meets, volunteered to teach at partner schools, managed social media, wrote blogs, and educational content on coding concepts.

ACADEMIC PROJECTS

Vigilance Vision: Developed a real-time crime detection system using OpenCV and FastAI to analyze CCTV footage and alert law enforcement of suspicious activity. Trained a ResNet50 model, achieving 98% accuracy in classifying potential threats.

Hate Speech Detection Application: Built a text classification web app using XGBoost & TF-IDF to detect & categorize hate/offensive speech with 93.7% accuracy & an F1 score of 0.9. Explored model interpretability using LIME & experimented with BERT & LSTM models.

Credit Risk Modelling: Designed a Flask-based application to predict loan eligibility, attaining 93% accuracy using CatBoostClassifier.

LEADERSHIP EXPERIENCE

Teaching Assistant, Data Algorithms Lab, MIT Manipal

Aug 2023 – Dec 2023

- Assisted with the Data Analytics and Algorithms Lab for 4th Semester students pursuing BTech Data Science by grading lab assignments and clearing doubts

General Secretary, Economics and Finance Society of Manipal

Aug 2022 – Nov 2023

- Orchestrated the execution of the club’s flagship event, Manipal Trading League, leading logistics and operations for 2000+ participants
- Increased club revenue by 48% by spearheading the IPL auction series, which engaged 100+ students in strategic mock auctions.
- Expanded outreach through 10+ finance, fintech, and ML workshops, and enhanced club accessibility by developing its official website

AWARDS AND ACHIEVEMENTS

- Patent pending**: System for integrating big data storage with quantum safe encryption technologies & method thereof, 2023
- Presented** Knock Knock: Through the Lens of Classification- Exploring State-of-the-art Models for Indoor Activity Recognition at ICCMEH’ 23
- 3rd Place, KAVACH Cyber Security Hackathon, GoI, 2023, and Top 30 (among 700 teams), TS NASSCOM Hackathon, India-wide, 2023

SKILLS AND INTERESTS

Programming Languages & Tools: Python, SQL, Perl, Go, R, C++, C#, Java, NoSQL, BashGit, Jupyter Notebook, PyArrow, DuckDB, PySpark

Technologies & Frameworks: Hadoop, Hive, Apache Spark, Kafka, Snowflake, HDFS, ETL Pipelines, MSSQL, Tableau, Splunk, Flink, Power BI

Professional Interests: Product Management, Data Engineering, Financial Technology, AI, Deep Learning, Analytics